Dear Editors,

I am submitting "Did the intervention change usage? Validity threats and difference-in-differences in before-and-after corpus designs" for consideration as a Research Article in *Research Methods in Applied Linguistics*.

Researchers across applied linguistics claim that a policy, a curriculum reform, a campaign or an editorial standard changed language use, and support the claim by comparing corpus frequencies before and after. The paper argues that this inference usually fails, and that the reason is a measurement problem before it is an estimation problem. A corpus frequency is a score. Like any score it stands in for a construct it does not equal, and the part of it contributed by the corpus's own composition and coding does not sit still: it moves systematically across periods for reasons unrelated to anyone's usage. A two-point comparison cannot separate that movement from an effect.

Difference-in-differences is the standard repair, and the paper translates its assumptions into corpus terms. The central result is that the design's identifying assumption has to hold twice, in the rate people use and in the corpus itself, and that the second is routinely assumed away. A numerical case shows the first holding while the second fails, producing a four-point effect where the true effect is zero. Readers who work with tests and scales will recognise the requirement: it is measurement invariance, moved from groups to periods and from levels to changes.

The paper then supplies what a reader can use: a reporting standard, a decision procedure with pre-specified gates and thresholds, and an explicit account of which verdicts those gates cannot reach. Simulations map where the procedure fails, including a false-positive rate that climbs to near certainty under composition drift the analyst cannot see. One composition margin of a real public archive is measured directly, to show that the confound is present and steep rather than hypothetical.

I am aware that the journal's readership works predominantly in second language acquisition, teaching and assessment, and that my worked material is diachronic and policy-facing. I have written the paper so that the argument arrives in validity terms rather than econometric ones, and so that the tools are named for a reader who works in R. The journal's scope statement lists corpus linguistics and language policy among the core areas of applied linguistics, and states that methods introduced from other disciplines are welcome where they are examined for the purpose of solving problems in this one. That is what I have tried to do.

Two disclosures. An earlier and substantially different version of this work is posted as a preprint at LingBuzz (lingbuzz/010080); it predates the present manuscript by several revisions. And I used large language models as drafting and editing aids, which is declared in the manuscript in the section and wording the journal specifies.

The manuscript is original, is not under consideration elsewhere, and has not been published. It reports no human-subjects research. Simulation and archive-query scripts are supplied so that every reported number can be reproduced.

Thank you for considering it.

Yours sincerely,

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